

This rubric explains how the quality of a student's Assignment 1 submission is evaluated.
For each criterion, markers select one level that best reflects the overall quality of submission and award a mark within the indicated range.

	Criteria	Developing	Nailing it	Solid work	Exceptional quality
SIT215 Assignment 1: Problem Solving – Rubric	Problem formulation, environment modelling, and completion (8)	- Identifies a problem scenario, but with limited clarity and minimal detail in the relevant modelling components. - Environment representation lacks essential features or clarity. Key constraints are missing or weakly linked to the problem. The PEAS framework is incomplete or used only descriptively without influencing modelling decisions. (1-2)	- The problem scenario and objective are clearly defined. The environment is modelled with identifiable and relevant components. - Environment representation includes key features and relevant constraints. Problem formulation and PEAS framework are clear and generally aligned with the objective. (3-4)	- The problem scenario and objective are clearly defined with modelling rationale, successfully translating them into specific and structured components. - Environment representation and PEAS framework are used analytically to justify modelling choices, with clearly defined constraints and features. Problem formulation is well-explained and fully connected to the task goal. (5-6)	- The problem scenario and objective are clearly defined with exceptional modelling rationale, with strong analytical insight and thoughtful abstraction. - Environment representation, PEAS framework, and abstraction is highly realistic and detailed, with constraints well integrated to reflect real-world challenges and their impact on search behaviour and feasibility. Problem formulation is comprehensive, insightful, and fully aligned with the task goal. (7-8)
	Algorithmic implementation competency, and comparison (12)	- Algorithms are partially implemented or poorly structured. Successor generation, goal testing, or cost tracking is unclear. - Explanation of algorithm choice or behaviour is limited. - Evaluation is mostly descriptive, with limited or unclear use of standard measurable metrics, resulting in weak or unclear comparisons. (1-3)	- Provides a functional solution meeting most requirements, with minor issues but a sound technical foundation. - The provided solution with A* and comparison algorithm(s) is generally structured and demonstrates a solid understanding of environment representation and search design. - Comparisons are explained clearly by using appropriate standard metrics. (4-6)	- Provides a fully functional solution, meeting all requirements as intended, demonstrating clear and effective structure. - The provided solution with A* and comparison algorithm(s) are correctly implemented and clearly explained, showing strong technical understanding of deterministic environment and search strategies. - Comparative evaluation is data-driven and well interpreted. Algorithm behaviour is analysed using standard search metrics and meaningful justification. (7-9)	- Demonstrates exceptional technical competency through a logically consistent and well-structured solution. - The solution with A* and comparison algorithm(s) is highly efficient, well-engineered, and robust, highlighting strong technical sophistication and mastery. - Evaluation demonstrates strong analytical thinking and effective use of performance evidence, with trade-offs critically discussed and explicitly links performance differences to constraint structure, heuristic, and intelligence quality, aligned with design decisions in the provided solution. (10-12)
	Application of theoretical concepts (12)	- Theoretical explanations are unclear or partially inaccurate. - Some inaccuracies or omissions limit understanding of heuristic function(s) and computational model(s). Heuristics are generally used but justification is weak, generic, or poorly connected to the environment and constraints. - Logical reasoning shows weak connection to the defined problem and solution. (1-3)	- Clearly explains and applies relevant theoretical concepts. - A* heuristic is appropriately justified based on task goals and model constraints. Any new heuristic design is appropriate and clearly explained. - Constraints and feasibility are mentioned but not clearly connected to theoretical reasoning. (4-6)	- Provides a detailed and accurate application of theoretical concepts. - Applies heuristics critically with data-driven justification. Theoretical choices are well explained and aligned with task goals, evaluation criteria, and the implemented solution. - Any new heuristic design reflects relevant environment structure and constraints, with strengths and limitations discussed using evidence from results. The feasibility discussion is addressed with clear logical reasoning. (7-9)	- Demonstrates advanced critical thinking by extending beyond standard theoretical concepts. - Applies computational models and heuristics with strong analytical depth, demonstrating comprehensive and relevant data-supported theoretical justification and evaluation. - Any new heuristic is insightful and logically well-reasoned. Trade-offs, admissibility, feasibility, and heuristic quality are critically examined with relevant logical reasoning. These are linked clearly to observed performance outcomes. (10-12)
	Reasoning, presentation, and communication (8)	- Report lacks consistent clarity or structure. Explanations are basic or fragmented. Code documentation and annotations are sparse or disconnected from the report. - Limited or ineffective use of appropriate visuals (where relevant). Inconsistent citation or referencing where applicable. (1-2)	- Able to present a clearly structured report as required. Code documentation and annotations are provided and mostly align with the report. - Provides appropriate visuals (where relevant), citations, and references where applicable. Minor issues with clarity or formatting. (3-4)	- Presents a professionally structured report. Code documentation and annotations are clear and closely aligned with the report explanations. - Provides effective use of appropriate visuals (where relevant), consistent formatting, data-driven analysis and comparison, and proper citations, reference where applicable. (5-6)	- Demonstrates outstanding presentation quality meeting professional standards. Code documentation and annotations, and report explanations are detailed, precise, and perfectly aligned. - Provides appropriate and high-quality visual aids (where relevant) that strongly support solution explanation and data-driven analysis. Citations and references are correctly formatted and consistently applied. (7-8)